Fiscal Year:	FY 2010	**Task Last Updated:**	FY 06/22/2010
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	Mapping gravitropic signaling and response: from ions to growth		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	05/27/2009	**End Date:**	05/26/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	June 2010: A postdoc has been hired to work on the project, Dr. Won-Gyu Choi who received hios Ph.D. in 2009 from the University of Tennessee.		
COI Name (Institution):	Masson, Patrick (University of Wisconsin-Madison) Spalding, Edgar (University of Wisconsin-Madison)		
Grant/Contract No.:	NNX09AK80G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research will address program priority 1 in the plant biology element of this NRA by seeking to define the intra-/inter-cellular mechanisms behind the generation and transmission of the gravity signal. It is widely accepted that sedimentation of starch filled amyloplasts in the columella of the root cap and endodermis of the hypocotyl act to initiate the gravitropic signal. However, although pH and Ca2+-based signals have been proposed as key elements in this plant response, the precise mechanisms translating physical force to biochemically transmissible signal is still poorly defined. Needed is a high resolution spatiotemporal map of changes in ion activities induced by the gravity signal. Such mapping will be achieved by confocal imaging of plants expressing a range of GFP-based probes targeted to different subcellular locales within the Arabidopsis seedling. In parallel, we will use machine-vision technologies to quantify the distribution of curvature along the root and hypocotyl axes during gravitropism. Curvature and ion change distribution maps will be measured in wild type and mutants known to be defective in the pathways that lead to gravity signal transduction. By mapping the detailed spatial and temporal dynamics of ion signaling to these morphometric analyses, we will be able to pinpoint where and when each component is altering tropic response. Potential for translation to future spaceflight opportunities: These studies will develop a morphometric technology highly applicable to spaceflight analysis where existing hardware can be used to collect the time-lapse imaging required for morphometric analysis. Such analysis will develop a quantitative description of the effects of the flight environment on plant development at levels of architecture ranging from branching patterns to the kinematics of growth. This will allow us to ask questions only addressable by the unique environment of Space, for example defining the effects of mutations on gravitational response thresholds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has capitalized on the development of image automated image analyses algorithms that allow the automated extraction of quantitative parameters of growth such as growth rate, organ angle, rate of curvature. Although these are being applied to root gravitropism in Arabidopsis, they hold the potential to be applicable to a much wider set of growth responses, allowing for the rapid screening of plants for quantitative differences in their growth patterns.
Task Progress:	This grant covers research to map the dynamics of ionic signaling associated with gravity sensing and response onto the kinetics of gravitropic growth to assess how well such signaling events can be linked to the control of plant development. We have concentrated on 2 of the major goals over the last year: (1) Developing a comprehensive catalog of gravitropic growth responses from the root systems of a range of mutants in Arabidopsis and (2) generating and optimizing the imaging tools for simultaneously monitoring ionic signaling events in wild type plants and in these mutant backgrounds. We have applied morphometric analysis of movies of tropic response in wild-type Arabidopsis from a range of ecotypes to define the natural variability in “wild-type” response and to develop a comprehensive description of wild-type response in the ecotypic backgrounds relevant for comparison to the gravitropism mutants under study (which are in a variety of ecotype backgrounds). We have then applied similar analysis to these mutants to define precisely where in time and space in their gravitropic response they exhibit alteration in growth control. Current analysis has captured approximately 230 trials, which we have determined is providing the robust dataset needed for statistical analysis and have captured the response kinetics of mutants ranging from the starchless pgm1 (which has a lesion in gravity sensing), to arg (which has a lesion in a likely component of the gravity sensor in the root cap), to aux1 (which is defective in auxin transport and so in auxin-driven gravitropic response). During this analysis we have also tested response under different growth conditions, concentrating on sucrose supplementation to the medium. This parameter is important as in the literature gravitropic response has been assayed under conditions of medium supplemented with sucrose ranging from 0 to 1% (0-33mM). As sucrose levels were increased in both wild-type and mutant plants, root growth rate also increased as did the rate of apparent gravitropic response. The morphometric analysis of root gravitropic curvature also captures incremental growth rate allowing us to correct the gravitropic response for changes in growth rate at each data point. This has allowed us to ask whether a slower gravitropic reorientation of the root in a particular mutant simply reflects a slower growth rate rather than an alteration in the kinetics of response more likely linked to a lesion in a fundamental aspect of gravitropic signaling or growth control. Such analysis has allowed us to define the mutants pgm1, aux1 and arg1 and mar2 in the arg1 background (mar1 and 2 are proteins found in the plastid membrane possibly playing a role in gravity sensing) as showing alterations in gravitropic response that cannot be wholly accounted for by their effects on growth rate. However, except for aux1 which remains essentially agravitropic under all sucrose levels, these other mutants show alterations in gravitropic response that are all significantly modulated by the sucrose level in the medium. Thus, for example, at 0 mM sucrose the pgm1 mutant shows agravitropic growth, but as sucrose levels are increased, its response begins to resemble wild-type plants until at 10 mM sucrose it is indistinguishable from wild-type. This morphometric analysis has provided us with a detailed map of growth responses in a range of gravitropic mutants onto which we can superimpose ionic signaling events (see below). The characterization of the sensitivity of gravitropic response to sucrose levels in the medium allows us to both compare our results to the those in the literature and to manipulate the tropic response between large kinetic differences to wild type – e.g. pgm1 under low sucrose – to response kinetics indistinguishable form wild type – e.g. pgm1 under 10 mM sucrose. Such manipulation is allowing us to ask if alterations in signaling activities seen under these conditions translate into an alteration in the kinetics of tropic response. i.e. altered signaling seen at 0 mM sucrose in pgm1 that is lost at 10 mM sucrose is more likely to be linked to growth rather than a fundamental lesion in gravity sensing per se as 10 mM sucrose rescues gravitropic growth response in pgm1 to wild-type levels. We have also been generating the ionic signaling probes to allow us to map changes in ion fluxes onto these growth kinetics. We have concentrated on 3 probes: the luminescent Ca2+ sensor aequorin, the fluorescent Ca2+ sensor YC3.6 and the fluorescent pH sensor GFP-H148D. These have been cloned and constructs developed with these proteins driven by promoters directing expression to either the whole plant, the root cap or the epidermis to ask if Ca2+ or pH changes in these regions of the plant occur as gravitropic signaling is initiated. At present these constructs are being transformed into wild-type plants and the gravitropic mutants outlined above.
Bibliography Type:	Description: (Last Updated: 02/02/2026)

Articles in Peer-reviewed Journals	Swanson SJ, Gilroy S. "ROS in plant development." Physiologia Plantarum. 2010 Apr;138(4):384-92. Review. PMID: 19947976 , Apr-2010